

VWAP Decision Framework

Pine Script v5 — TradingView Indicator Suite

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Overview

This Pine Script indicator suite implements the VWAP-based decision framework documented in the Agentic Decision Architecture research. It adds three analytical panels to any TradingView chart, providing real-time signal detection, momentum measurement, and entry/exit alerts based on the validated three-criteria entry framework.

Three Indicator Panels

Panel	Function	Colour Coding
Panel 1	VWAP Slope — 10-minute rate of change of the VWAP middle line. Identifies session character: steep decline, moderate decline, flat, or rising.	Red = steep decline. Orange = moderate. Grey = flat. Green = rising.
Panel 2	VWAP % Deviation — price distance from VWAP expressed as a percentage. More meaningful than absolute dollar deviation as VWAP moves through the session.	Blue = oversold zone. Red = overbought zone. Teal = below VWAP. Grey = above VWAP.
Panel 3	Entry/Exit Signal — fires +1 when all three entry criteria are simultaneously met (zone, RSI, BEP). Fires -1 when exit conditions are detected.	Green histogram = entry signal. Red histogram = exit signal. Background highlights on signal bars.

Automated Alerts

Four alerts are built into the script:

1. Entry Signal — fires when all three criteria (zone, RSI below 40, full BEP) are simultaneously confirmed
2. Exit Signal — fires when RSI reaches overbought or bearish engulfing is detected
3. Zone Entry — fires when price first enters the -1SD to -2SD VWAP bounce zone
4. Steep VWAP Decline — fires when slope crosses the steep decline threshold, signalling conservative exit targets

Installation

1. Open TradingView and navigate to your chart
2. Click Pine Editor at the bottom of the screen
3. Paste the full script below and click Save
4. Click Add to Chart
5. Calibrate thresholds in the Settings panel once live data is available

Calibration Note

The default threshold values in this script are starting points. After adding the indicator to your chart, calibrate the following inputs against your own validated dataset before trading:

- VWAP Slope thresholds (slope_steep, slope_moderate, slope_flat) — align to your bounce size data
- VWAP % Deviation levels — calibrate to where -1SD and -2SD typically sit as a percentage on your instrument
- RSI threshold — default 40, do not change without data supporting a different level

Pine Script Source Code

Copy the complete code below and paste into TradingView Pine Editor:

```
// =====
// INTC Day Trading Framework – VWAP Indicators
// Developed for: Dermot | AWS ASEAN AI & Data GTM Lead
// Platform: TradingView Pine Script v5
// Instrument: INTC (NASDAQ) | Timeframe: 1-minute
// =====
// INSTRUCTIONS:
// 1. Open TradingView and navigate to INTC 1-minute chart
// 2. Click Pine Editor at the bottom of the screen
// 3. Paste this entire script and click Save
// 4. Click Add to Chart
// 5. This script adds THREE panels to your chart:
//     Panel 1: VWAP Slope (10-minute rate of change)
//     Panel 2: VWAP % Deviation (price distance from VWAP as %)
//     Panel 3: Entry Signal Alert (all three criteria simultaneously)
// =====

//@version=5
indicator(title="INTC Trading Framework – VWAP Indicators",
          shorttitle="INTC Framework",
          overlay=false,
          max_labels_count=500)

// =====
// INPUTS – Adjust these to calibrate to your validated thresholds
// =====

// VWAP Slope Settings
slope_lookback = input.int(10, title="VWAP Slope Lookback (minutes)", minval=5, maxval=30,
group="VWAP Slope")
slope_steep    = input.float(-0.25, title="Steep Decline Threshold", group="VWAP Slope")
slope_moderate = input.float(-0.10, title="Moderate Decline Threshold", group="VWAP Slope")
slope_flat     = input.float(0.05, title="Flat/Rising Threshold", group="VWAP Slope")

// RSI Settings
rsi_length     = input.int(14, title="RSI Length", group="RSI")
rsi_oversold   = input.float(40.0, title="RSI Oversold Threshold", group="RSI")
rsi_overbought = input.float(70.0, title="RSI Overbought Threshold", group="RSI")

// VWAP % Deviation Settings
dev_oversold   = input.float(-1.5, title="% Deviation Oversold Level", group="VWAP % Deviation")
dev_overbought = input.float(1.5, title="% Deviation Overbought Level", group="VWAP % Deviation")

// Display Settings
show_slope     = input.bool(true, title="Show VWAP Slope Panel", group="Display")
show_dev       = input.bool(true, title="Show VWAP % Deviation Panel", group="Display")
show_signal    = input.bool(true, title="Show Entry Signal Panel", group="Display")
show_alerts    = input.bool(true, title="Enable Alerts", group="Display")

// =====
// CALCULATIONS
// =====
```

```

// --- VWAP ---
// Uses session-anchored VWAP with HLC/3 source (standard institutional setting)
vwap_val = ta.vwap(hlc3)

// --- VWAP Slope ---
// Rate of change of VWAP over the lookback period
// Negative = declining VWAP session, Positive = rising VWAP session
vwap_slope = vwap_val - vwap_val[slope_lookback]

// --- VWAP % Deviation ---
// How far price is from VWAP expressed as a percentage
// More meaningful than absolute dollar deviation as VWAP moves through session
vwap_pct_dev = ((close - vwap_val) / vwap_val) * 100

// --- RSI ---
rsi_val = ta.rsi(close, rsi_length)

// --- VWAP Bands ---
// Standard deviation bands - matches TradingView VWAP indicator settings
vwap_stdev = ta.stdev(close, slope_lookback)
upper_1sd = vwap_val + vwap_stdev
lower_1sd = vwap_val - vwap_stdev
upper_2sd = vwap_val + (2 * vwap_stdev)
lower_2sd = vwap_val - (2 * vwap_stdev)

// --- Zone Detection ---
// TRUE when price is between -1SD and -2SD (the bounce entry zone)
in_bounce_zone = close < lower_1sd and close > lower_2sd

// --- Candle Type ---
is_red_candle = close < open
is_green_candle = close > open

// --- RSI Condition ---
// RSI below oversold threshold on a red candle - Criterion 2
rsi_oversold_condition = rsi_val < rsi_oversold and is_red_candle

// --- Bullish Engulfing Detection ---
// Checks if current green candle fully engulfs prior red candle body
// Green open <= Prior red close AND Green close >= Prior red open
prev_is_red = close[1] < open[1]
engulf_open_ok = is_green_candle and prev_is_red and open <= close[1]
engulf_close_ok = is_green_candle and prev_is_red and close >= open[1]
is_engulfing = engulf_open_ok and engulf_close_ok

// --- Combined Entry Signal ---
// All three criteria simultaneously:
// 1. Price was in zone on prior red candle
// 2. RSI was below 40 on prior red candle
// 3. Current candle is a full bullish engulfing
prior_zone = close[1] < lower_1sd[1]
prior_rsi_ok = rsi_val[1] < rsi_oversold
entry_signal = is_engulfing and prior_zone and prior_rsi_ok

// --- Exit Signal ---
// RSI overbought OR bearish engulfing at top
bearish_engulf = is_red_candle and close[1] > open[1] and open >= close[1] and close <= open[1]
exit_signal = rsi_val > rsi_overbought or bearish_engulf

// =====
// PANEL 1 - VWAP SLOPE
// =====

// Colour coding:
// Red (dark) = Steep decline (worse than -$0.25 over 10 min) - conservative target
// Orange = Moderate decline (-$0.10 to -$0.25) - standard target
// Gray = Flat (-$0.10 to +$0.05) - baseline $0.84 target
// Green = Rising (above +$0.05) - extended target, hold toward VWAP

slope_color = vwap_slope < slope_steep ? color.new(color.red, 20) :
              vwap_slope < slope_moderate ? color.new(color.orange, 20) :
              vwap_slope < slope_flat ? color.new(color.gray, 20) :
              color.new(color.green, 20)

slope_target_label = vwap_slope < slope_steep ? "STEEP - Target $0.50-0.75" :
                    vwap_slope < slope_moderate ? "MODERATE - Target $0.75-1.50" :

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        vwap_slope < slope_flat      ? "FLAT – Target $1.50-2.00"      :
        "RISING – Hold toward VWAP"

// Plot slope as histogram
plot(show_slope ? vwap_slope : na,
     title="VWAP Slope (10-min)",
     style=plot.style_histogram,
     color=slope_color,
     linewidth=2)

// Reference lines for slope thresholds
hline(slope_steep,    "Steep Threshold",    color=color.new(color.red, 50),
      linestyle=hline.style_dashed)
hline(slope_moderate, "Moderate Threshold", color=color.new(color.orange, 50),
      linestyle=hline.style_dashed)
hline(0,             "Zero Line",           color=color.new(color.white, 50),
      linestyle=hline.style_solid)
hline(slope_flat,    "Rising Threshold",    color=color.new(color.green, 50),
      linestyle=hline.style_dashed)

// =====
// PANEL 2 – VWAP % DEVIATION
// =====

// Shows how far price is from VWAP as a percentage
// Negative = price below VWAP (potential bounce zone)
// The -1SD and -2SD zone roughly corresponds to -1% to -2% deviation on INTC
// (calibrate dev_oversold and dev_overbought inputs to match your actual band levels)

dev_color = vwap_pct_dev < dev_oversold ? color.new(color.blue, 20) :
            vwap_pct_dev > dev_overbought ? color.new(color.red, 20) :
            vwap_pct_dev < 0             ? color.new(color.teal, 40) :
            color.new(color.gray, 40)

plot(show_dev ? vwap_pct_dev : na,
     title="VWAP % Deviation",
     style=plot.style_line,
     color=dev_color,
     linewidth=2)

hline(dev_oversold, "Oversold Zone", color=color.new(color.blue, 40),
      linestyle=hline.style_dashed)
hline(0,           "VWAP Level",    color=color.new(color.white, 50),
      linestyle=hline.style_solid)
hline(dev_overbought, "Overbought Zone", color=color.new(color.red, 40),
      linestyle=hline.style_dashed)

// =====
// PANEL 3 – ENTRY AND EXIT SIGNALS
// =====

// Plots a simple signal line:
// +1 = Valid entry signal (all three criteria met)
// 0 = Neutral
// -1 = Exit signal (RSI overbought or bearish engulfing)

signal_value = entry_signal ? 1.0 : exit_signal ? -1.0 : 0.0

signal_color = entry_signal ? color.new(color.green, 0) :
               exit_signal ? color.new(color.red, 0) :
               color.new(color.gray, 80)

plot(show_signal ? signal_value : na,
     title="Entry/Exit Signal",
     style=plot.style_histogram,
     color=signal_color,
     linewidth=3)

hline(1, "Entry Signal", color=color.new(color.green, 60), linestyle=hline.style_dashed)
hline(0, "Neutral",     color=color.new(color.white, 60),  linestyle=hline.style_solid)
hline(-1, "Exit Signal", color=color.new(color.red, 60),  linestyle=hline.style_dashed)

// =====
// LABELS – Show key information at signal points
// =====

// Entry signal label

```

```

if entry_signal and show_signal
    label.new(bar_index, 1.2,
        "ENTRY SIGNAL\nZone ✓ RSI ✓ Engulf ✓\nSlope: " +
str.tostring(math.round(vwap_slope, 2)) + "\nRSI: " + str.tostring(math.round(rsi_val, 1)),
        color=color.new(color.green, 20),
        textcolor=color.white,
        style=label.style_label_down,
        size=size.small)

// Exit signal label
if exit_signal and show_signal
    label.new(bar_index, -1.2,
        "EXIT SIGNAL\nRSI: " + str.tostring(math.round(rsi_val, 1)),
        color=color.new(color.red, 20),
        textcolor=color.white,
        style=label.style_label_up,
        size=size.small)

// =====
// ALERTS
// =====

if show_alerts
    // Entry alert – fires when all three criteria are simultaneously met
    alertcondition(entry_signal,
        title="INTC Entry Signal",
        message="INTC ENTRY SIGNAL – All three criteria confirmed. Zone ✓ RSI ✓
Full Engulfing ✓. Check TradingView before executing on Saxo.")

    // Exit alert – fires when overbought or bearish engulfing appears
    alertcondition(exit_signal,
        title="INTC Exit Signal",
        message="INTC EXIT SIGNAL – RSI overbought or bearish engulfing detected.
Review exit target.")

    // Zone entry alert – fires when price first enters the bounce zone
    alertcondition(in_bounce_zone and not in_bounce_zone[1],
        title="INTC Zone Entry",
        message="INTC – Price has entered the -1SD to -2SD bounce zone. Watch for
RSI below 40 and bullish engulfing.")

    // Steep VWAP decline alert
    alertcondition(vwap_slope < slope_steep and vwap_slope[1] >= slope_steep,
        title="INTC Steep VWAP Decline",
        message="INTC – VWAP slope has entered steep decline territory. Reduce
profit targets if entering a bounce trade.")

// =====
// BACKGROUND COLOURING (optional visual aid)
// =====

// Highlight the background when an entry signal fires
bgcolor(entry_signal ? color.new(color.green, 85) : na, title="Entry Signal Background")
bgcolor(exit_signal ? color.new(color.red, 85) : na, title="Exit Signal Background")

// =====
// NOTES FOR CALIBRATION
// =====
// After adding this indicator to your chart, calibrate the following:
//
// 1. VWAP Slope Thresholds (slope_steep, slope_moderate, slope_flat)
//    Run your 60-day analysis and find the actual slope values that correlate
//    with small vs large bounces. Replace the default values with your findings.
//
// 2. VWAP % Deviation Levels (dev_oversold, dev_overbought)
//    Check where -1SD and -2SD correspond to in percentage terms on INTC.
//    Typically -1SD is around -0.8% to -1.2% and -2SD is around -1.5% to -2.0%
//    but this varies by session volatility. Calibrate from your data.
//
// 3. RSI Threshold (rsi_oversold)
//    Currently set to 40 (your validated moderate threshold).
//    Do not change without data supporting a different level.
//
// 4. The engulfing detection in Pine Script is approximate –
//    always verify the exact open/close prices manually on TradingView
//    before executing. Pine Script processes in real time and the signal
//    should be treated as an alert to look, not an automatic trade trigger.

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// =====

Part of the Agentic Decision Architecture research series. Full framework available at dermotgreally.github.io